2024312506 余沅桦 6一时变电磁场

1.解：由于 $l \gg r_1$ $l \gg r_2$，所以两柱型极板间的场可以看成是无限长带电圆柱面产生的场，设柱型极板上电荷线密度为 τ，选取柱型高斯面，由高斯定理可得：

$\oint_S \vec{D} \cdot d\vec{S} = D \cdot 2\pi r \cdot l = \tau l$ 解得 $D = \frac{\tau}{2\pi r}$ $E = \frac{\tau}{2\pi\varepsilon r}$

两极板之间的电势差为

$u = \int \vec{E} \cdot d\vec{l} = \int_{r_1}^{r_2} \frac{\tau}{2\pi\varepsilon r} dr = \frac{\tau}{2\pi\varepsilon} \ln\frac{r_2}{r_1} = \frac{\tau}{2\pi\varepsilon} \ln 4$

解得 $\tau = \frac{2\pi\varepsilon u}{\ln 4}$ $E = \frac{u}{r \ln 4}$

则 $D = \frac{\varepsilon}{r \ln 4} 6000\sqrt{2} \sin 100\pi t$ $J = \frac{4\varepsilon_0}{r \ln 4} 6000\sqrt{2} \cdot 100\pi \cos 100\pi t$

2.解：设球形电容器带有电量为 q 由高斯定律

$\oint_S \vec{D} \cdot d\vec{S} = D 4\pi r^2 = q$ 解得 $D = \frac{q}{4\pi r^2}$ $E = \frac{q}{4\pi\varepsilon r^2}$

$u = \int \vec{E} \cdot d\vec{l} = \int_a^b \frac{q}{4\pi\varepsilon r^2} dr = \frac{q(b-a)}{4\pi\varepsilon ab}$

解得 $q = \frac{4\pi\varepsilon ab u}{b-a}$ $D = \frac{q}{4\pi r^2} = \frac{\varepsilon ab (U_m \sin\omega t)}{r^2 (b-a)}$

$J_D = \frac{\partial D}{\partial t} = \frac{\varepsilon ab \omega (U_m \cos\omega t)}{r^2 (b-a)}$

$J_C = \gamma E = \frac{\gamma ab (U_m \sin\omega t)}{r^2 (b-a)}$

则全电流 $I = (J_C + J_D) 4\pi r^2 = \frac{4\pi ab U_m}{b-a} (\gamma \sin\omega t + \varepsilon\omega \cos\omega t)$

3.解：由麦克斯韦第二方程可知 $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$$\nabla \times \vec{E} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & 30\pi \cos(\omega t - \frac{4}{3}y) \end{vmatrix} = -30\pi \sin(\omega t - \frac{4}{3}y)(-\frac{4}{3}) \vec{e}_x$$

解得 $\vec{B} = -\int 40\pi \sin(\omega t - \frac{4}{3}y) \vec{e}_x dt = \frac{40\pi}{\omega} \cos(\omega t - \frac{4}{3}y)$

$\vec{H} = \frac{40\pi}{\mu_r \mu_0 \omega} \cos(\omega t - \frac{4}{3}y) \vec{e}_x$ A/m

$\vec{H} = 10 \cos(\omega t - \frac{4}{3}y) \vec{e}_x$ A/m 解得 $\omega = 10^7$ rad/s

由麦克斯韦第一方程可知，非导电媒质中 $(\gamma = 0)$

$\nabla \times \vec{H} = \vec{J}_C + \frac{\partial \vec{D}}{\partial t} = \frac{\partial \vec{D}}{\partial t}$

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$$\nabla \times \vec{H} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 10\cos(\omega t - \frac{4}{3}y)\vec{e}_x & 0 & 0 \end{vmatrix} = -\frac{40}{3}\sin(\omega t - \frac{4}{3}y)\vec{e}_z$$

解得 $\frac{\partial \vec{E}}{\partial t} = -\frac{40}{3}\sin(\omega t - \frac{4}{3}y)\vec{e}_z$

$\vec{D} = -\int \frac{40}{3}\sin(\omega t - \frac{4}{3}y)\vec{e}_z\,dt = \frac{40}{3\omega}\cos(\omega t - \frac{4}{3}y)\vec{e}_z$

$\vec{E} = \frac{\vec{D}}{\varepsilon_r \varepsilon_0} = \frac{40}{3\varepsilon_r \varepsilon_0 \omega}\cos(\omega t - \frac{4}{3}y)\vec{e}_z$ V/m

又 $\vec{E} = 30\pi\cos(\omega t - \frac{4}{3}y)\vec{e}_z$ V/m

解得 $\varepsilon_r = 1600$

4. 解：由已知得 $E_{1n} = E_1\cos 45° = 50\sqrt{2}\sin(100t + 30°)$

$E_{1t} = E_1\sin 45° = 50\sqrt{2}\sin(100t + 30°)$

由衔接条件可知 $E_{1t} = E_{2t}$

$E_{2t} = 50\sqrt{2}\sin(100t + 30°)$

干土导电率 $\gamma_2 = 10^{-5}$ S/m 可近似为理想介质

则 $D_{2n} - D_{1n} = 0$ $E_{2n} = \frac{\varepsilon_0 E_{1n}}{\varepsilon_{r2}\varepsilon_0} = \frac{50\sqrt{2}}{3}\sin(100t + 30°)$

所以 $E_2 = \sqrt{E_{2t}^2 + E_{2n}^2} = 74.54\sin(100t + 30°)$ V/m

$\alpha_2 = \arctan\frac{E_{2t}}{E_{2n}} = 71.57°$

5. 解：分界面法向方向为 $\vec{e}_n = \vec{e}_y$ $\vec{K} = \vec{e}_n \times \vec{H} = \vec{e}_y \times \vec{H}$

因为理想导体内电场为零 $= H_0\sin\beta x\cos(\omega t - \beta y)\vec{e}_y$

$E_{2n} = E_{2t} = 0$

所以分界面处 $\vec{E}$ 的切向分量为零

解：1. $\vec{S} = \vec{E} \times \vec{H} = 2.12 \times 10^4 \sin(\omega t - \beta z)\vec{e}_z$ W/m²

2. $\vec{S_{av}} = \mathrm{Re}[\vec{E} \times \vec{H}] = \mathrm{Re}[2000 e^{-j\beta z}\vec{e}_x \times 5.3 e^{j\beta z}\vec{e}_y]$

所以 $\vec{S_{av}} = 1.06 \times 10^4 \vec{e}_z$ W/m²

当 $z = 0$ 时，流入大面积体的瞬时功率为 P_1

$P_1 = 2.12 \times 10^4 \sin^2(\omega t) \times 0.5$

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当 z=2 时，流出六面体的瞬时功率为 P_2

$P_2 = 2.12 \times 10^{4} \sin(\omega t - 2\beta) \times 0.5$

流入六面体的净瞬时功率为

$P = P_1 - P_2 = 29 \times 10^{3} \sin(2\omega t - 0.84)\ W$

7. 解：坡印亭定理的数学表达式为：

$$-\int_V \frac{\partial w}{\partial t} dV = \oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} + \int_V (\vec{E} \cdot \vec{J}) dV$$

$-\int_V \frac{\partial w}{\partial t} dV$：单位时间内电容器中电磁能量的减少量

$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$：单位时间内透过 s 流出电容器的电磁能

$\int_V (\vec{E} \cdot \vec{J}) dV$：电容器内单位时间的热损耗

(2) $\vec{E} = \frac{V_0}{d} \vec{e}_z$　$\vec{J} = \gamma \vec{E} = \frac{\gamma V_0}{d} \vec{e}_z$　$\oint_L \vec{H} \cdot d\vec{l} = \sum I$

$H(2\pi\rho) = J\pi\rho^2 = \frac{\gamma V_0}{d} \pi \rho^2$　$\vec{H} = \frac{\gamma \rho V_0}{2d} \vec{e}_\phi$

$\vec{S} = \vec{E} \times \vec{H} = \frac{\gamma \rho V_0^2}{2d} (-\vec{e}_\rho)$

(3) $W_e = \int_V \frac{1}{2} \vec{D} \cdot \vec{E} dV = \frac{1}{2} \frac{\varepsilon \pi a^2}{d} V_0^2$

$\oint_A \vec{S} \cdot d\vec{S} = -\frac{\gamma \pi a^2}{d} V_0^2$

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